Skylark Labs - Full Stack Engineer Coding Test

Project: RTSP Stream Viewer

Overview

Build a simple web application that allows users to add RTSP stream URLs and view the live streams in a browser. The application should be deployed on Vercel, Render, Railway or a similar service.

The stack for this test is **TypeScript** and **Go (Golang)**.

Required Deliverables

- GitHub repository with complete source code
- Live demo hosted on Vercel, Render, Railway or a similar platform
- README with setup instructions

Technical Requirements

1. Frontend

- Create a clean, responsive web interface with:
    - Input field to add RTSP stream URLs
    - Display area for the live stream(s)
    - Option to view multiple streams simultaneously in a grid layout
    - Basic stream controls (play/pause)
- Use React.js for the frontend implementation

2. Backend

- Create a backend that:
    - Accepts RTSP stream URLs
    - Processes video streams using FFmpeg
    - Streams video to the frontend over WebSockets
    - Handles connection errors gracefully

3. Deployment

- Host the frontend on Vercel, Render, Railway or a similar service
- Deploy the backend on a platform of your choice (Vercel, Render, Railway, etc.)
- Provide clear instructions for running the application locally

Evaluation Criteria

- Functionality: Streams display properly and smoothly
- Code quality and organization
- Error handling
- UI/UX design
- Performance with multiple streams
- System architecture and scalability

Test RTSP Streams

For testing purposes, set up your own RTSP streams using MediaMTX and FFmpeg.

Submission Instructions

1. Send us the GitHub repository URL
2. Provide the URL to the live demo
3. Include any credentials needed to access the application (if applicable)
4. Timeline 7 days from the receipt of this test

Good luck!